Quantum Computing

Part 3 - Hardware by Modality: Neutral Atom Qubits

Compiled by the Spaced Research Ledger

Last updated: 2026-08-30

Neutral atom machines hold uncharged atoms in place with focused laser beams called optical tweezers, and make them interact by exciting them into large, puffed-up states known as Rydberg states. The atoms are identical by nature, like the ions in Part 2. They are not electrically charged, so they can be packed far more densely without pushing each other apart.

That density is the modality's defining advantage. The systems in this part reach thousands of qubits while the machines in Parts 1 and 2 count in the tens and low hundreds. The offsetting weakness is that atoms escape their traps, so a large array steadily loses its qubits, and gate fidelities sit slightly behind the leaders in the other two modalities.

A second distinction matters for reading the numbers below. These machines run in two modes. Analog mode excites all atoms at once to simulate a physical system directly. Digital mode executes ordinary quantum gates one after another for general algorithms [2]. A large analog machine and a large digital one are not the same achievement.

Section 1 covers the academic groups setting the records, Section 2 QuEra, Section 3 Pasqal, Section 4 Infleqtion, and Section 5 Atom Computing.

1. Academic Records

The largest arrays and the highest fidelities in this part come from university groups rather than companies. Two results here also address the atom-loss problem directly, which is the constraint that governs how large these machines can get.

Recent Developments

The Harvard group led by Mikhail Lukin reached 99.85% raw fidelity for its two-qubit gates, rising to 99.94% when results affected by atom loss are discarded [1]. The gap between those two numbers is the atom-loss penalty stated plainly.

Caltech's Endres group published the first direct experimental measurement of two-dimensional conformal field theory spectra, using an analog simulator with chains of up to 35 strontium atoms [4]. This is a physics result obtained on a quantum machine rather than a machine result, and it is the kind of work analog mode exists for.

Current State

The Harvard group, with collaborators including Hollerith at ETH Zurich, reported continuous operation of a coherent system of more than 3,000 qubits [2]. It used a dual optical-lattice conveyor belt that continuously moves atoms from a reservoir into the computing zone. Tweezers load at 300,000 atoms per second.

Continuous loading is the answer to atom loss. Rather than trying to stop atoms escaping, the machine replaces them faster than they leave. That array was sustained for over two hours, which QuEra and its Harvard collaborators presented as resolving the atom-loss problem for scaling [3].

Caltech holds the size record. In September 2025 the Endres group reported an array of 6,100 caesium atoms trapped across 12,000 optical tweezer sites [2]. Single-qubit manipulation accuracy was 99.98%, and the group demonstrated coherent atom transport. The work was published in Nature as a tweezer array with 6,100 highly coherent atomic qubits [5]. The group has also published a benchmarking and fidelity-response theory for high-fidelity Rydberg entangling gates in PRX Quantum, establishing how to characterize two-qubit gate errors on its platform [5].

2. QuEra

QuEra's machines are the most used neutral-atom systems in the world by customer hours, and its gate fidelities are the strongest of the commercial platforms in this part.

Recent Developments

QuEra's Gemini system reached 99.77% two-qubit gate fidelity, rising to 99.96% when atom-loss cases are excluded, and averaging 99.74% across large zones [6].

Current State

The result that established the modality's credibility came earlier. QuEra, with Harvard and MIT, achieved 99.5% two-qubit entangling fidelity on up to 60 rubidium atoms operating in parallel, published in Nature in October 2023 [7]. That figure passed the surface-code error-correction threshold, meaning the hardware became good enough for the error-correction schemes in Part 6 to help rather than hurt.

QuEra's Aquila is a 256-atom analog processor available on Amazon Braket since 2022 [8]. It is the most-accessed neutral-atom system in the world measured by external customer hours, which is a different kind of achievement from any fidelity figure.

3. Pasqal

Pasqal has the widest commercial deployment in this part and the most unusual strategy. It ships analog and digital processors at the same time and upgrades one hardware line toward fault tolerance, rather than treating today's machines as disposable.

Recent Developments

Pasqal had 7 processors with 100 to 200 or more qubits deployed in commercial use as of August 2026, with 3 more in production [9]. Deployment count is the number that distinguishes this company from its competitors.

Pasqal and its subsidiary Aeponyx demonstrated trapping individual atoms using laser light generated by a photonic integrated circuit [10]. The company announced it in August 2026 as a world first. Putting qubit control onto a chip addresses the practical problem that these machines currently need a table of optical equipment.

CNRS, the Institut d'Optique Graduate School and Pasqal opened a joint laboratory called AtomIQ-Lab in November 2025 [14]. It focuses on laser-controlled cold-atom computers and on improving processor performance through atom-manipulation techniques.

Current State

Pasqal's array quality result is the strongest evidence that this modality can scale cleanly. It achieved defect-free 1,024-atom registers with a 0.3% average defect rate [11]. Atom lifetimes ran beyond 5,000 seconds in a 4-kelvin cryogenic environment. Long lifetimes attack the same atom-loss problem that Section 1 solves by continuous replacement.

Deployment has been steady. Pasqal put its 100-qubit Orion Beta system on site at the French GENCI computing centre and the German Jülich Research Centre between 2024 and 2025 [2]. These were among the first quantum computers integrated into supercomputing facilities. In February 2026 it delivered a system of more than 140 qubits to the Italian CINECA centre and launched its Vela processor with more than 256 qubits [2]. It had announced a processor with more than 1,000 atoms loaded in a single event in June 2024 [2].

One claim needs a caveat. In early 2026 Pasqal announced a demonstrated quantum advantage over classical methods in a materials-science simulation of magnetic materials [12]. This is a company statement that has not been peer-reviewed, published, or independently confirmed.

The company's strategy is to ship analog and digital processors concurrently rather than separating current machines from future fault-tolerant ones, upgrading the same platform over time [13].

4. Infleqtion

Infleqtion competes on installed national infrastructure rather than raw scale. Its systems are the operational machines at the UK and Japanese national programmes, and its recent work extends to using two different atomic species at once.

Recent Developments

Japan's Institute for Molecular Science, with Hitachi and Infleqtion, launched Shunkai, Japan's first full-stack neutral-atom quantum computer [15]. It starts at 50 qubits with a target of 10,000 by 2031, and uses rubidium atoms and optical tweezers. It runs at room temperature with reconfigurable connectivity. Cloud access is planned, with commercialization through Yaqumo Inc.

Infleqtion reported a record dual-species entangling gate fidelity of 0.975 ± 0.002, using rubidium and caesium together, in May 2026 [16]. Two species allow one type of atom to be measured while the other keeps its state, which matters for error correction.

Current State

Infleqtion demonstrated arrays up to 1,600 sites with 99.73% two-qubit gate fidelity on its Sqale platform [17]. It had reported that same fidelity in December 2025 alongside 114-qubit in-place entanglement, then the largest UK neutral-atom array [18].

The UK installation is the company's anchor. In December 2025 the Sqale system at the UK National Quantum Computing Centre became the country's only operational 100-qubit machine [19]. That followed a February 2025 demonstration of a 256-site array at the Centre's Harwell campus, the largest of its kind in the UK at the time [21].

The Sqale architecture combines individual optical addressing of qubits with dynamically reconfigurable arrays [20]. It supports both atom rearrangement and in-place entanglement in a digital gate-based mode.

5. Atom Computing

Atom Computing runs the largest production machine in this part and stores information differently from everything else here, in nuclear spins rather than electronic states. That choice buys coherence times measured in seconds.

Current State

The Phoenix system holds 1,180 physical qubits in a 1,225-site optical tweezer array [8]. Announced in late 2023, it was the first universal quantum platform to pass 1,000 qubits, and as of 2026 it remains the largest production neutral-atom system.

The coherence figure is the standout. Atom Computing encodes qubits in the nuclear spin states of strontium atoms, which yields coherence times on the order of 40 seconds [22]. Every other machine in this collection measures coherence in microseconds or milliseconds.

The company operates a gate-based digital architecture rather than a purely analog one [22]. It has reported 99.6% two-qubit gate fidelity, described as the highest for any commercial neutral-atom platform [22].

Unresolved Conflicts

Two companies claim the highest commercial two-qubit fidelity in this modality, and both cannot be right. Atom Computing reports 99.6% and describes it as the highest for any commercial neutral-atom platform [22]. Infleqtion reports 99.73% on its Sqale platform, a higher figure billed the same way [17]. The Atom Computing source is undated and low confidence, so it cannot be established which claim is current. A dated measurement from either company, stating the array size and whether atom-loss cases were excluded, would settle it.

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

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